Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	PQC Hardware Security Module (HSM) Solution with SoC-Based Optimized Post-Quantum Cryptography			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity (√), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum (√)			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information and Communication		Electrical and Electronic Engineering	
	Medium Category	Small Category	Medium Category	Small Category
	Cybersecurity	General Security Technology	Semiconductor Components and Systems	SoC
Concept of Proposed Technology	This project is an international industry-academia collaborative R&D program that develops defense and industrial solutions by implementing cryptographic modules optimized for NIST-standardized post-quantum cryptography (PQC) algorithms, specifically ML-KEM and ML-DSA, on FPGA platforms, integrated with university academic research for real-world deployment. The primary target application is high-performance network Hardware Security Module (HSM) environments, which require high throughput, scalability, and long-term operational stability as core requirements. The goal is to develop a product-level FPGA-based PQC-HSM solution that			

	achieves balanced optimization in terms of performance, hardware efficiency, and security robustness. Through this approach, the project aims to bridge the gap between academic novelty and practical product implementation. Rather than being limited to single algorithm implementation or simple performance improvements, this project proposes next-generation cryptographic hardware optimization technology that aligns with the post-standardization phase of PQC by simultaneously pursuing academic novelty and industrial applicability.
Rationale for Joint R&D Project	Post-quantum cryptography hardware implementation is a highly challenging field that requires optimized design and verification infrastructure. By combining the fundamental algorithm and hardware optimization capabilities of Canadian universities with the application and commercialization expertise of domestic defense and security equipment companies, synergy can be maximized toward productization. Additionally, collaborative research with overseas university research institutions enables the development of globally competitive reference implementations and establishes a foundation for international market penetration in alignment with NIST standards and PQC transition policies across different countries.
Key Words	1. Post-Quantum Cryptography (PQC) 2. Hardware Cryptographic Accelerator 3. FPGA Optimization Design 4. HSM (Hardware Security Module)

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3-4)	End of the Project	Level (6)
R&D Objectives	This project develops a product-level FPGA-based PQC-HSM solution applicable to defense and industrial network environments, while simultaneously expanding academic knowledge on FPGA optimization and hardware design for side-channel attack countermeasures. **Academic Objectives** 1. **PQC Hardware Optimization:** Optimization of ML-KEM/ML-DSA operations (achieving **performance/area/energy efficiency considerations**) 2. **Security Verification:** Establishing side-channel attack resistance for post-quantum cryptography **Industrial Objectives** 1. **PQC-HSM Integration Development:** Development of optimized PQC-integrated HSM architecture 2. **Multi-Session PQC Performance:** Ensuring PQC performance in multi-session environments 3. **PQC-HSM Demonstration:** Integration and demonstration of PQC-HSM solution-applied platform prototype 4. **Performance, Stability, and Security assurance:** KCMVP validation compliance level (preliminary review)			
R&D Target Specifications	• **Quantitative Objectives** - ML-KEM performance targets (performance/area metrics to be established) - ML-DSA performance targets (performance/area metrics to be established) - Side-channel countermeasure targets (feasible criteria to be established) - KCMVP cryptographic module preliminary review (FIPS-140 compliance) - Demonstration (testbed) – (1-2 actual solution replacement items) • **Qualitative Objectives** - Achieving domestic leading performance in ML-KEM (key exchange algorithm) (requires revision) - Achieving domestic leading performance in ML-DSA (digital signature algorithm) (requires revision) - Development of PQC-enabled KCMVP cryptographic module applicable to various industrial sectors - Development of high-performance PQC-HSM solution prototype			
R&D Plan	**Year 1: PQC-HSM Architecture Design and FPGA Optimization and Verification of PQC Algorithms** • PQC-HSM requirements analysis and architecture design • ML-KEM / ML-DSA hardware optimization design and development - NTT operation pipeline architecture optimization - SHA3/SHAKE hardware optimization			

(max 10 pages)

	• Trade-off analysis of PQC cryptographic IP for area/performance/energy optimization
	
	Year 2: PQC-HSM Development and Side-Channel Countermeasure Optimization Design for PQC Algorithms
	• Development of KCMVP validation-ready PQC cryptographic module and HSM
	  – PKCS#11 interface API development
	  – Cryptographic module development integrating PQC crypto IP
	• ML-KEM / ML-DSA hardware side-channel analysis and countermeasure technology development
	• FPGA-based PQC-HSM prototype development
	
	Year 3: Side-Channel Protected PQC Integration and PQC-HSM Enhancement
	• Establishment of real test-bed environment for HSM with integrated PQC algorithms
	• Analysis and verification of FPGA-applied side-channel countermeasure technology and PQC cryptographic module enhancement
	• KCMVP validation preliminary review for PQC-integrated cryptographic module

(max 10 pages)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• **Test Procedure** – Collaboration with SME partner for finalizing the items to be tested • **ML-KEM/ML-DSA Hardware Optimization Design and Enhancement** – Collaboration with Korean SME Partner for PQC cryptographic algorithm communication interface design – NTT (Number Theoretic Transform) computational pipeline architecture optimization – SHA3/SHAKE hardware accelerator optimization • **PQC Cryptographic IP Area/Performance Trade-off Analysis** – FPGA resource utilization analysis (Slice, LUT, DSP, BRAM) – Frequency-area-power trade-off simulation • **ML-KEM/ML-DSA Hardware Side-Channel Analysis** – Power analysis attack (DPA, CPA) vulnerability assessment – Timing attack vulnerability assessment • **Side-Channel Countermeasure Development and Enhancement** – Analysis of masking/shuffling/constant-time side-channel countermeasure techniques – Side-channel resistant RTL design and verification • **Technology Transfer and Academic Achievements**
Expected R&R of Korean SME Partner	• **Test Procedure** – Collaboration with SME partner for finalizing the items to be tested • **PQC-HSM Requirements Definition and Architecture Design** – FPGA-based HSM hardware platform development – Key management system design (Key Lifecycle Management) – Network interface design (TCP/IP, TLS) • **PQC Cryptographic IP-Integrated Cryptographic Module Development** – FPGA board selection and PCB design – Collaboration with Research Institute for Crypto IP ↔ HSM interface protocol design – Development of cryptographic module with integrated PQC crypto IP • **FPGA-Based PQC-HSM Prototype Development** – PKCS#11 standard interface implementation – HSM architecture design and development with cryptographic module integration – Hybrid mode support with legacy cryptography (RSA, ECDSA) – Network stack integration development (TCP/IP, TLS) • **KCMVP Validation of PQC Cryptographic IP-Integrated Cryptographic Module** • **PQC-HSM Hardware Prototype Fabrication**

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	• **PQC-HSM-Based Solution Testbed Demonstration**
Test and Evaluation Center	• Proposing preliminary test items based on similar evaluations • Collaboration with the university and SME partner for finalizing the items to be tested

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technical Expected Outcomes** • Proactive response to quantum-resistant cryptography transition technology through hybrid/PQC implementation capabilities • Complete compatibility with existing HSM ecosystems through PKCS#11 standard compliance • Integrated implementation of masking, shuffling, and constant-time techniques at the FPGA level to ensure protection against power analysis attacks (DPA/CPA) and timing attacks • Establishment of domestic and international certification pathways through full compliance with NIST FIPS 203/204 standards and successful KCMVP preliminary review, providing a foundation for meeting international security requirements including FIPS-140 **Academic Expected Outcomes** • Development of generalized principles for memory hierarchy optimization, pipeline stage partitioning, and parallel processing design for NTT operations in lattice-based cryptography, presenting a standard reference framework applicable to various lattice-based cryptographic schemes • Positioning Korea as a leader in PQC hardware technology through publication in SCI-level international journals such as IEEE TIFS and ACM TECS, and presentations at top-tier hardware security conferences including CHES and DAC • Training of hardware security engineers and PQC/FPGA security specialists, and securing next-generation cryptographic hardware experts at a global level through personnel exchange with Canadian universities **Industrial and Policy Expected Outcomes** • Acceleration of commercialization for defense and industrial PQC-HSM security products in the era of quantum-resistant cryptography transition • Establishment of a foundation for entry into North American and European quantum-resistant cryptography markets by leveraging NIST standards-based technology secured through joint research with Canadian universities • PQC hardware technology and certification experience serve as a demonstration of the national quantum-resistant cryptography transition roadmap • Strengthening of the domestic defense SME ecosystem through establishment of domestic production systems for PQC-integrated security modules, and regional economic revitalization through creation of new employment opportunities for specialized personnel in hardware design, verification, and manufacturing sectors

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6. References

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(3) NIST Post-Quantum Cryptography Standardization Project (2016-2024)

(4) Oder et al., "Practical NTT-Based Polynomial Multiplication on FPGA" (CHES 2016)

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(9) AMD Xilinx, "UltraScale+ FPGA Architecture and Product Data Sheet" (2023)

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(11) Kearney et al., "High-Throughput Cryptographic Acceleration on FPGA: Architectural Considerations" (ACM FPGA 2019)

(12) ETSI, "Quantum-Safe Cryptography and Security: Industry Readiness Analysis" (2022)

(13) Thales Group, "Luna Network HSM – Architecture and Performance Overview" (2023)

(14) Amazon Web Services, "AWS CloudHSM: Architecture and Performance Characteristics" (2023)

(15) FIPS 140-3: Security Requirements for Cryptographic Modules, Available: https://csrc.nist.gov/pubs/fips/140-3/final

(max 10 pages)

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization